

Introduction to Cryptography and Security

Exercise 15: Certificates and PKI

January 5, 2026

Authentication protocol

- Bob $\rightarrow$ Alice: n_B
- Alice $\rightarrow$ Bob: $n_A \parallel \text{sig}_A(n_A \parallel n_B \parallel ID_B)$
- Bob $\rightarrow$ Alice: $n'_B \parallel \text{sig}_B(n'_B \parallel n_A \parallel ID_A)$

- 1 Describe the objectives of each step.
- 2 Explain the interleaving attack shown in the problem and its consequences. Compare this to a man-in-the-middle attack.

Needham-Schroeder Public Key (NSPK)

- ① $A \rightarrow B : \{n_A, A\}_{pk_B}$
- ② $B \rightarrow A : \{n_A, n_B\}_{pk_A}$
- ③ $A \rightarrow B : \{n_B\}_{pk_B}$

Assume an intruder **I** is a legitimate user in the network. Alice initiates a session with **I**, but **I** wants to impersonate Alice to Bob.

Show the sequence of messages for a Man-in-the-Middle (MITM) attack where Bob believes he is talking to Alice, but **I** possesses the secret nonce n_B .

Active attack against Diffie–Hellman with certificates:

- 1 Oscar replaces $Cert(B)$ with a valid $Cert(O)$. How is this detected?
- 2 Oscar replaces b_B with his own public key b_O . How is this detected?

Assume Oscar gets hold of the CA's private key. Can he decrypt old ciphertexts exchanged before the compromise? If not, what other attack can he perform?

Oscar replaces the CA's public key with his own on Bob's system.

- 1 Describe how Oscar handles Bob's certificate request so Bob doesn't find out.
- 2 Describe how Oscar establishes a session key with Bob such that Bob thinks he is talking to Alice.